

In [2]:

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import pandas as pd
import numpy as np
import yfinance as yf
import matplotlib.pyplot as plt

# Download historical stock data
ticker = 'AAPL'
data = yf.download(ticker, start='2020-01-01', end='2024-07-22')

# Calculate RSI
def calculate_rsi(data, window=14):
    delta = data['Close'].diff()
    gain = (delta.where(delta > 0, 0)).rolling(window=window).mean()
    loss = (-delta.where(delta < 0, 0)).rolling(window=window).mean()
    rs = gain / loss
    rsi = 100 - (100 / (1 + rs))
    return rsi

data['RSI'] = calculate_rsi(data)

# Implement the strategy
def implement_rsi_strategy(data):
    buy_signals = []
    sell_signals = []
    position = False # False means no position, True means holding a stock

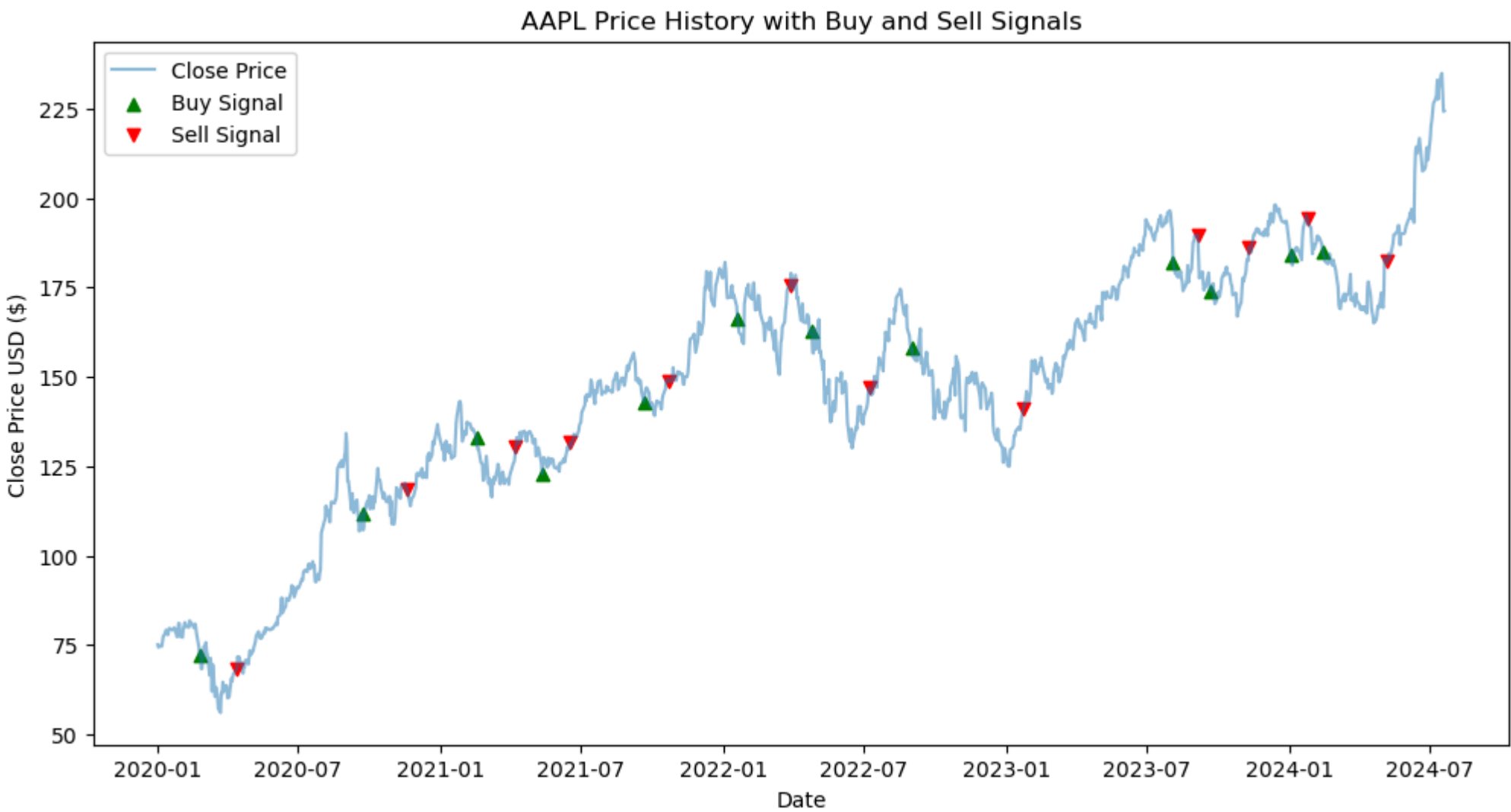
    for i in range(len(data)):
        if data['RSI'].iloc[i] < 30 and not position:
            buy_signals.append(data['Close'].iloc[i])
            sell_signals.append(np.nan)
            position = True
        elif data['RSI'].iloc[i] > 70 and position:
            buy_signals.append(np.nan)
            sell_signals.append(data['Close'].iloc[i])
            position = False
        else:
            buy_signals.append(np.nan)
            sell_signals.append(np.nan)

    return buy_signals, sell_signals

data['Buy_Signal_Price'], data['Sell_Signal_Price'] = implement_rsi_strategy(data)

# Plot the results
plt.figure(figsize=(12,6))
plt.plot(data['Close'], label='Close Price', alpha=0.5)
plt.scatter(data.index, data['Buy_Signal_Price'], label='Buy Signal', marker='^', color='green', alpha=1)
plt.scatter(data.index, data['Sell_Signal_Price'], label='Sell Signal', marker='v', color='red', alpha=1)
plt.title(f'{ticker} Price History with Buy and Sell Signals')
plt.xlabel('Date')
plt.ylabel('Close Price USD ($)')
plt.legend(loc='best')
plt.show()
```

[*****100%*****] 1 of 1 completed



In []: